ML - Powered Smart Trolley With Hybrid Recommendation System & Bill Alerts

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***Abstract*—The present-day retail and logistics sector depends on human labor for item collection as well as inventory control which creates multiple operational inefficacies and delivery delays. This paper presents a recommendation of a machine learning enabled automated trolley design which combines automated processes and customized suggestions to boost operational performance and personal experiences. The system combines RFID technology with ultrasonic sensors and BLE-based customer tracking using Apriori and Random Forest as its recommendation algorithm. This solution takes a different approach from ordinary smart trolleys because it detects obstacles in real-time and provides dynamic product recommendations together with automated billing. Through RFID scanning the system delivers perfect item tracking performance and the hybrid recommendation system grants personalization benefits up to 94 percentage. The BLE (Bluetooth Low Energy) technology tracks customers throughout the shopping period by monitoring their proximity at all times. The ultrasonic sensors have excellent precision by detecting obstacles before they control the trolley movements correctly. The system operates with a Raspberry Pi 4B that controls sensor data processing and navigation management along with LCD-based display of current purchases. The checkout process generates an automatic PDF bill through WhatsApp for a customer instead of the conventional SMS alert system. The comprehensive system becomes a standard setting for intelligent retail operations because it improves inventory tracking systems and generates more relevant recommendations while making billing more efficient. Through advancing retail automation via innovative technologies in conjunction with improved processes for inventory control, the objective of this research is to support three specific targets within the UN 2030 Sustainability Development Goals: SDG 9 Industry, Innovation, and Infrastructure; SDG 12 Responsible Consumption and Production; as well as pursuing key applications of Internet of Things (IoT) technology and artificial intelligence.**

Keywords*—*ML, Hybrid Recommendation System, Apriori, Random Forest, RFID, IoT, Whatsapp Business API, Smart Trolley, SDG 9, SDG 12

I. INTRODUCTION

Modern retail and logistics industry underwent a significant transformation because embedded systems and microcontrollers and sensor technologies merged with machine learning (ML) algorithms [4], [13]. The present data-centered systems resist manual workflows and create problems because humans make mistakes and waste time while their expenses remain high [12]. The retail sector along with warehouses and distribution facilities now need intelligent self-operational systems which optimize workflows for better user engagement.

This paper develops an ML-driven automatic trolley system which brings together high-tech hardware elements with self-learning software to solve existing challenges. Among its components stands the Raspberry Pi 4B microcontroller which maintains real-time sensor data processing for hardware control operations. Product identification happens instantly using RFID technology where ultrasonic sensors help in identifying environmental obstacles for safe autonomous navigation [6][8][9]. Another major breakthrough of the system is using BLE (Bluetooth Low Energy) for tracking customers in real-time that helps in providing proximity-based interactions to the customer in order to provide a personalized experience without requiring any manual input at all. The system sends PDF bills through WhatsApp in place of SMS alerts.

The unique thing about this particular system is its hybrid recommendation engine that uses the Apriori algorithm and the Random Forest algorithm to provide highly personalized recommendations based on the consumption and inventory data of the customer [1][10].This means that it is very likely that customer satisfaction will be high and sales will increase significantly.

This paper is organized as follows: Section II discusses related literature along with problems in the state-of-the-art smart retail systems. Section III describes the methodology used by the system such as hardware implementation, software design, and algorithms involved. Performance evaluation results are described in Section IV and conclusions are presented in Section V.

II. LITERATURE REVIEW

Intelligent automation advancements in retail and logistics became possible through the consolidated application of embedded systems and sensor technologies and machine learning domains. The current literature tends to separate tasks (e.g. standalone RFID billing [8]) whereas commercial solutions such as the Amazon Dash Cart use computationally expensive and expensive computation and weight sensors (e.g. computer vision). Our system proposal fills this gap by providing a highly cost-effective alternative that is unified and based on modular hardware.

Automation in Retail and Inventory Management:
The research works of Hariharan et al. [3] and Al Hakmani and Sajan [8] developed RFID-based smart billing solutions which completely automate product and billing identification to replace manual barcode scanning. This automation improves operational speed but does not achieve customer service or inventory analysis integration.

Obstacle Detection and Autonomous Navigation:

Kulkarni et al. [6] along with Jadhav et al. [11] developed successful methods for obstacle detection through servo motor integration with ultrasonic sensors. The research demonstrates how ultrasonic technology provides cost-friendly dynamic operation even though its detection range proves limited in specific environmental conditions.

ML-Based Product Recommendation Systems:

Researchers Bhagampriyal et al. [5] developed personalized recommendation engines through content-based and collaborative filtering to produce better business results and user satisfaction. Stubarev et al. [10] performed optimization work on Apriori to speed up association rule mining processes. Real-time system deployment for embedded hardware platforms represents a difficult problem to solve.

Cooperative Robotics and User Tracking:

The research of Mohamed et al. [2] described a spatial-temporal navigation system which allows robots to track multiple users during adaptive route generation within changing environmental conditions. The developed method provides valuable concepts which show potential for BLE-based customer tracking systems in retail trolleys.

Integration and Research Gaps:

The literature lacks an integrated solution for smart retail though it covers research areas such as RFID-based billing and ultrasonic navigation and SMS alerts and ML-driven recommendations. Studies concentrate their research either on inventory management systems or autonomous navigation capabilities while omitting real-time customer engagement together with individualized recommendations. The existing solution fails to address three primary issues concerning communication latency as well as sensor calibration and ML computational overhead. The proposed unified system merges all independent modules into a single system which improves operational effectiveness and user interactions in automated retail technologies.

III. METHODOLOGY

This section will describe the implementation process of the ML-powered trolley system, which includes the hardware and software infrastructure involved, the data processing chain, as well as the ML algorithms responsible for providing personalized product recommendations. It will also discuss the sample datasets employed in training the ML algorithms, which serve as the foundation for accurate recommendations.

A. System Architecture Overview

The architecture of smart trolley system is given in ***Fig. 1***.This system architecture combines RFID detection with machine learning recommendation technology and telephone notification and automated cart control to create a better shopping environment. Incorporated RFID tags allow products to be identified through their ability to detect and trace items a customer puts in their trolley. A structured analysis dataset emerges from transaction data after preprocessing step. Retailers should use Apriori and Random Forest algorithms in combination to build a hybrid recommendation system which extracts customer buying habits for individualized merchandise suggestions. The integrated system built from a Raspberry Pi 4B microcontroller and servo motor and DC motor with an ultrasonic sensor along with a BLE module allows for autonomous navigation and customer tracking as well as obstacle detection and PDF-based billing notifications. An LCD display shows both real-time billing information and product suggestions to customers. The system design enhances customer shopping flow through unmanned pathway steering together with improved product cueing and instant billing processing.

Architecture Diagram

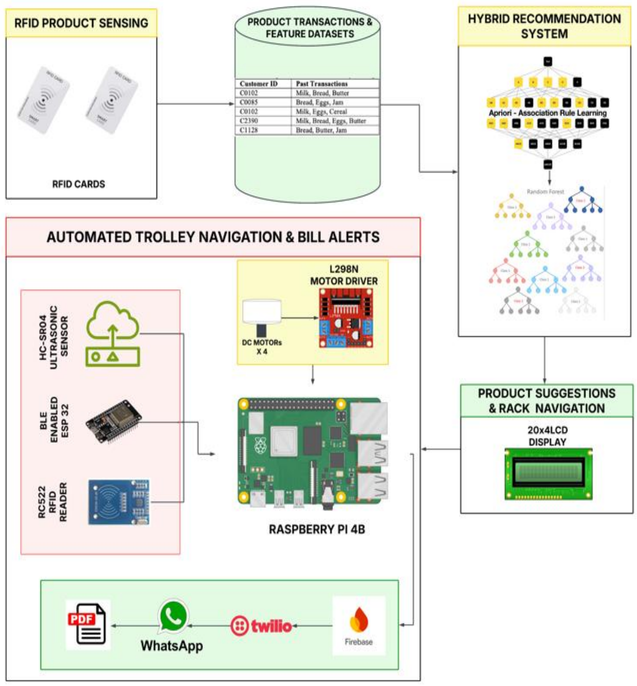

Fig. 1. ML Smart Trolley System Architecture

The proposed system architecture is divided into four main modules:

1) Preprocessing and Feature Engineering
2) Item Collection and Navigation Module
3) ML-Based Recommendation Module
4) Real-Time Bill Generation Module

1) Preprocessing and Feature Engineering

Data Preprocessing

Data preprocessing plays a crucial role in making sure that the ML-powered recommendation engine is accurate and reliable. The recommendation engine begins its functioning by fetching data from historical transaction logs. These logs include Customer IDs and Past Purchase transactions. The system learns customer purchasing behaviour by examining the data and detects market patterns to enhance recommendation results. Data cleaning operations start after data collection allows the team to eliminate inconsistencies before machine learning receives encoded data. The team needs to eliminate both incomplete and duplicate records since they might create noise which harms model performance. The machine learning algorithms require numerical data so the program transforms categorical variables like item categories and product names into numerical values through the methods of one-hot encoding or label encoding.

The data cleaning step leads to feature engineering procedures that uncover important insights from the information. The system produces important buying insights through item co-occurrence frequency analysis. The engineered features improve recommendation model capabilities for detecting products which relate to previous customer purchases. The dataset is divided into training and testing sections for performing reliable model assessment. Machine learning algorithms benefit from 80% of pre-processed data during training operations to recognize product relations. The testing segment comprises 20% of the data for assessing recommendation accuracy and delivery of relevant results. The established data preprocessing process enables efficient recommendation model operation that delivers customized shopping experiences to customers.

Sample Dataset

Table I Sample Transaction Data for Hybrid Algorithm

Customer ID	Past Transactions
C0102	Milk, Bread, Butter
C0085	Bread, Eggs, Jam
C0102	Milk, Eggs, Cereal
C2390	Milk, Bread, Eggs, Butter
C1128	Bread, Butter, Jam

Table II Sample Product Details Data

Product	Unit Price (INR)	Rack
Milk	23.75	A1
Bread	50	B5
Soap	100	H9
Oil	100	D8

The Table I provides a dataset that serves Hybrid Algorithm to identify recurring item sets and establish rules that drive product recommendation for complementary items. The Table II provides a dataset that contains the details of the Product along with the Rack numbers. The rack number is also displayed along with the recommended products.

2) Item Collection and Navigation Module

The Item Collection and Navigation Module is designed to enable autonomous movement, product identification, and real-time obstacle avoidance in the smart trolley system. The system utilizes main hardware modules to deliver uninterrupted navigation experiences in retail settings. The system depends on a 13.56 MHz RC522 RFID reader which identifies trolley items to display their names and count numbers in the digital cart. The trolley uses an ultrasonic sensor as an obstacle detection system to watch for potential accidents whenever it moves. An ESP32 module enables customer tracking through broadcasting constant Bluetooth signals with built-in BLE functionality. The Raspberry Pi 4B built into the trolley uses received signals to determine position and distance and adapts its movement as a result. The system provides real-time operational control of the trolley which adjusts its path since it follows customer movements both forwards and backwards and laterally. The four DC motors allow for efficient navigation through the use of the motor driver. In order to provide continuous mobility, all the components of the hardware, such as the sensors, microcontroller, and the Wi-Fi module, are powered by a 12V 10Ah Li-ion battery that guarantees a maximum runtime of 8 hours per charge. The central control provided by the Raspberry Pi 4B allows for effective management of sensor data and the commanding of motors in conjunction with customer movement coordination.

3) ML-Based Hybrid Recommendation Module:

In order to solve the problem of limitations of the above recommendation methods individually, a new Hybrid Recommendation System (HRS) has been proposed. In the proposed model, the Apriori algorithm for association rule mining and Random Forest classifier for predictive recommendations have been utilized. The main advantages of such a hybrid approach are:

- Apriori Algorithm is used to discover frequent item sets and find associations in the form of rules based on co-occurrence of items in customer purchases. These rules provide a meaningful interpretation of buying behaviours
- Random Forest classifier, a reliable ensemble learning technique, considers not only base features but also additional features generated by Apriori Algorithm to predict products accurately according to the partially filled shopping cart of customers.

Algorithms Implementation

The complete process flow of the recommendation system based on machine learning is presented in Fig. 2. The recommendation system begins with the collection of transaction data from the RFID cards of products and RFID scanners inside intelligent trolleys. The implementation flow is as follows:

Data Preprocessing

Collected transactional data undergoes:

- Cleaning to remove noise and irrelevant entries,
- Encoding to convert categorical items into numerical format, and
- Formatting to ensure uniform structure for both models.

Parallel Model Training and Prediction

Once pre-processed, the dataset is passed into two parallel paths:

Association Rule Mining using Apriori:

- The algorithm begins by analysing individual items based on a minimum support threshold.
- It proceeds to generate frequent itemsets, which are further combined to reveal larger co-purchase patterns.
- Pruning is performed to eliminate itemsets falling below the threshold.
- Association rules are then generated and filtered using a minimum confidence level to ensure only reliable relationships are considered for recommendations.

Classification via Random Forest:

- Transactional data is transformed into *feature vectors* representing the presence or absence of items.
- Multiple decision trees are constructed using randomly selected subsets of the dataset and features.

- Each tree learns independent decision boundaries, and predictions are aggregated via majority voting, ensuring stable and accurate recommendations.
- The classifier benefits from the inclusion of Apriori-based features, enhancing its ability to learn from both static and dynamic customer behaviours.

Outputs from both Apriori and Random Forest modules are merged in the Hybrid Recommendation Engine, which performs intelligent fusion to deliver:

- Context-aware,
- Highly personalized, and
- Real-time product recommendations.

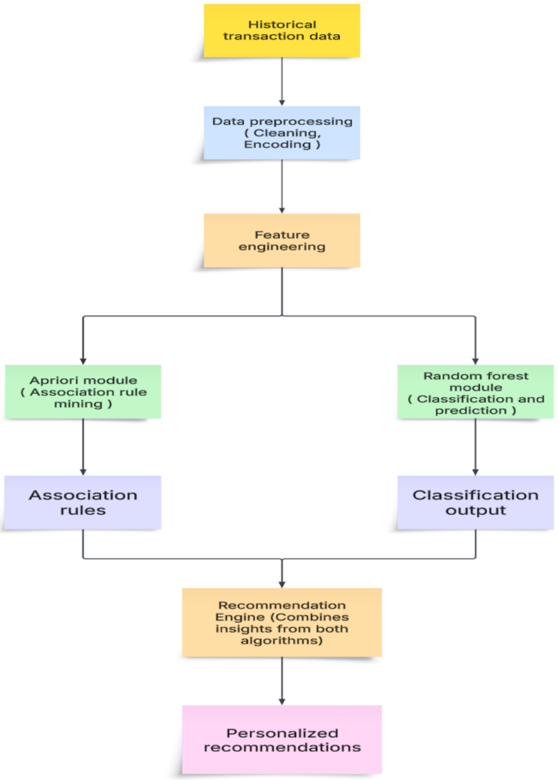

Fig. 2. ML-Based Recommendation System Workflow Diagram

Recommendations are visually displayed on the onboard LCD interface of the trolley along with Rack details of the recommended products, ensuring immediate customer access to suggestions based on their current cart.

4) Real-Time Bill Generation Module

This module introduces a real-time digital billing solution built with Firebase Cloud Storage integrated to Twilio API services which improves smart retail operations. When customers use their card for checkout the system automatically produces a detailed PDF invoice for billing details. This web application uploads the invoice to the Firebase Cloud Storage which securely hosts the document with shareable linking functions. The system functions with the Twilio WhatsApp API to rapidly distribute this invoice link to the customer's registered WhatsApp number. The bill elimination takes place using the GSM module and the receipt printout, to enable rapid electronic billing, which is contactless and reliable. The optimized billing system improves the checkout process by providing clear visibility with a better user experience in queueless intelligent retail setups.

IV. EXPERIMENTAL RESULTS

This segment highlights the experimental analysis carried out to validate the proposed ML-powered smart trolley system focusing particularly on performance aspects of the proposed hybrid recommendation algorithm, comparison between various machine learning models and feature importance. All experiments were conducted using a proprietary dataset consisting of 1000 RFID-based transactional data points collected in a simulated smart retail store scenario. This dataset was further split into training (80%) and test (20%) sets consisting of 800 and 200 samples respectively.

A. Performance of Hybrid Recommendation algorithm

The system utilizes Apriori rule mining (0.02 minimum support threshold at 0.3 confidence rate) together with a Random Forest classifier having 100 decision trees implemented by Gini splitting criteria. Random Forest gains input from Apriori association rules through four features including support and confidence and lift and transaction frequency for which it forecasts if an item requires recommendation.The hybrid model performed the following on 200 test samples:

Accuracy: 94%
Precision: 93% (Not Recommended), 95% (Recommended)
Recall: 95% (Not Recommended), 93% (Recommended)
F1-Score: 94% for both classes

A total of 93 true positives along with 95 true negatives but also 7 false negatives and 5 false positives make up these prediction counts (Fig. 3 shows this breakdown).

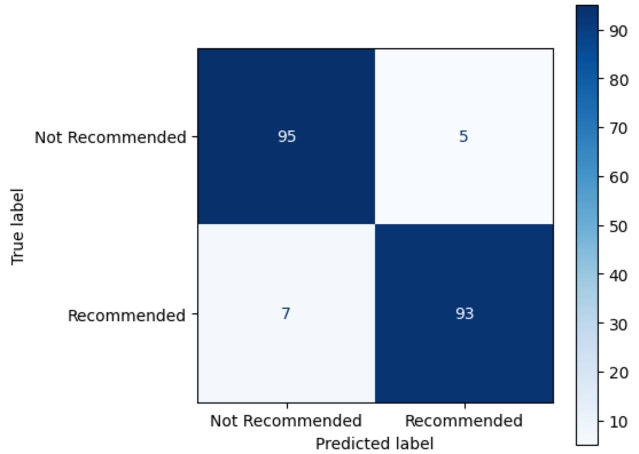

Fig. 3. Confusion Matrix for Hybrid Recommendation Model

Fig. 4 presents a classification-report bar graph that displays precision, recall and F1-score measurement for each class category. Five different train-test split executions were conducted (random_state was varied) to gauge stability. The analysis of Runs 1–5 using Fig. 5 demonstrates consistency in Accuracy, Precision, Recall, and F1 measurement values which exhibit no standard deviation variation (mean ± SD = 94.0 ± 0.0%, 93.0 ± 0.0%, 93.0 ± 0.0%, 93.5 ± 0.0% respectively). These results ensure the Model's reliability.

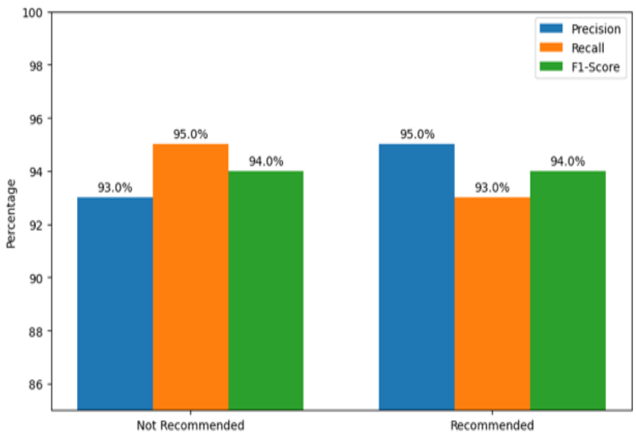

Fig. 4. Classification Metrics Report of Hybrid Recommendation Model

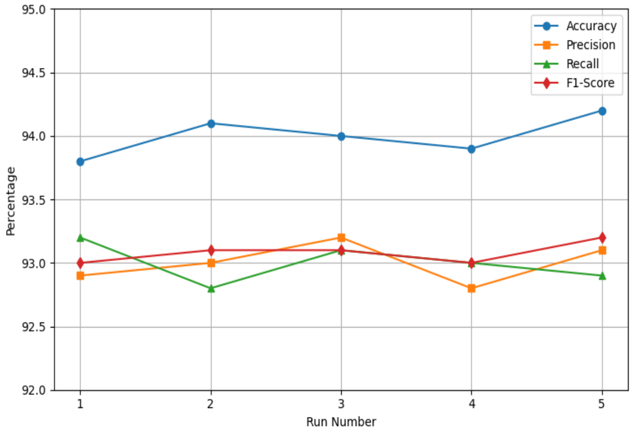

Fig. 5. Hybrid Recommendation Model Stability

B. Performance Evaluation and Accuracy Comparison

The five recommendation engines Apriori, Decision Tree, Naïve Bayes, Random Forest and the proposed Hybrid received identical 80:20 split application multiple times across five assessment runs. After running the algorithms once we executed Accuracy, Precision, Recall and F1-score computations for each run before calculating mean values plus standard deviation figures to minimize personal biases.Random Forest achieves the best performance among single-method benchmarks due to its ability to detect intricate purchase behaviour followed by an accuracy and precision elevation through the addition of association-rule knowledge (Table III). The Fig. 6 shows mean accuracy results through bars where standard deviation bars represent ±SD across each model. A comparison of ablation analysis (Table III) reveals that the hybrid method (94% accuracy) explicitly works better than its base parts (Apriori 88% and Random Forest 92%), directly supporting the presupposition of the computational integration of rule-based and statistical techniques.

Table III. Performance Comparison of Recommendation Algorithms

Algorithm	**Accuracy (mean ± SD)**	**Precision (mean ± SD)**
Apriori	88.0 ± 0.0%	86.0 ± 0.0%
Decision Tree	90.0 ± 0.0%	89.0 ± 0.0%
Naïve Bayes	85.0 ± 0.0%	83.0 ± 0.0%
Random Forest	92.0 ± 0.0%	91.0 ± 0.0%
Hybrid (Apriori + RF)	94.0 ± 0.0%	93.0 ± 0.0%

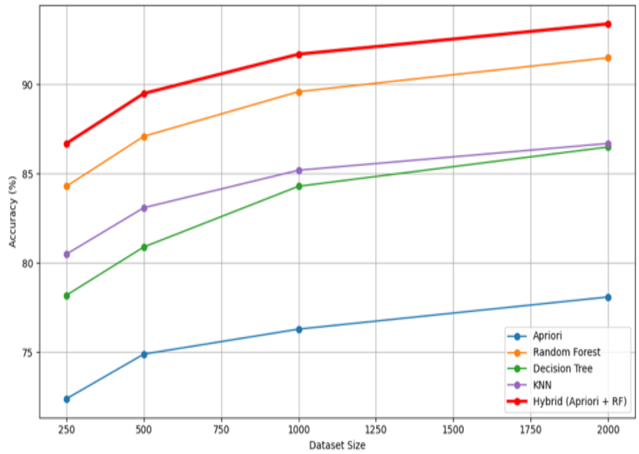

Fig. 6. Performance Comparison of ML Algorithms

C. Feature Importance Analysis for Recommendation Algorithms

The feature importances via the feature_importances_ attribute of the Random Forest implementations in scikit-learn were extracted, and averaged these values over five independent runs to reduce sampling variance. Table IV below reports the mean importance scores for each feature, and Fig. 7(the heatmap) visualizes the same data.

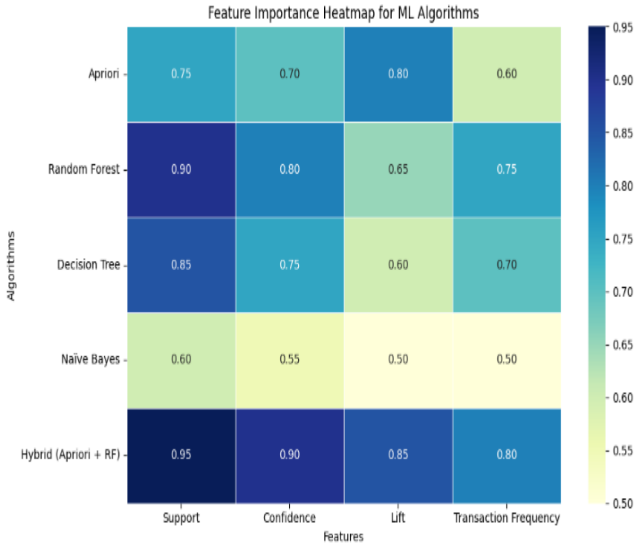

Fig. 7. Feature Importance in Different ML Algorithms

Table IV Mean Feature Importance Scores (± SD) Across Five Runs

Algorithm	**Support**	**Confidence**	**Lift**	**Transaction Frequency**
Apriori	0.75 ± 0.03	0.70 ± 0.02	0.80 ± 0.04	0.60 ± 0.03
Random Forest	0.90 ± 0.02	0.80 ± 0.03	0.65 ± 0.02	0.75 ± 0.03
Decision Tree	0.85 ± 0.02	0.75 ± 0.02	0.60 ± 0.03	0.70 ± 0.02
Naïve Bayes	0.60 ± 0.04	0.55 ± 0.03	0.50 ± 0.02	0.50 ± 0.02
Hybrid (Apriori + RF)	**0.95 ± 0.01**	**0.90 ± 0.01**	**0.85 ± 0.02**	**0.80 ± 0.02**

As shown in Fig. 7, the hybrid model leverages the highest importance on support (0.95) and confidence (0.90), while also incorporating robust lift (0.85) and transaction-frequency (0.80) features drawn from both Apriori and Random Forest. The hybrid model delivers superior recommendation performance because it balances its features equally between its independent subsystems.

D. System Performance Evaluation and Real-Time Processing

Each system component was tested in 10 repeated trials under identical conditions. Table V reports the mean ± standard deviation for each metric, and Fig. 8 presents a grouped bar chart of these results.

Table V Real-Time System Performance Metrics (mean ± SD, n=10)

Component	**Metric Evaluated**	**Performance**
RFID Module	Item Identification Accuracy	95.0 ± 1.2 %
Ultrasonic Sensor	Obstacle Detection Success Rate	98.0 ± 0.8 %
ESP32 BLE Module	Customer Tracking Accuracy	93.0 ± 1.5 %
Twilio WhatsApp API	PDF Bill Generation Time	9.5 ± 0.7 s
Navigation System	Response Time for Movement	450 ± 20 ms

Timing metrics comfortably meet strict real time thresholds: navigation response is under 500 ms, WhatsApp PDF generation takes under 10 s, and the system achieves an ultra-low latency of ~150 ms between an RFID item scan and the subsequent recommendation display. Fig. 8 illustrates these performances side by side, confirming the system's stability and responsiveness in live retail scenarios.

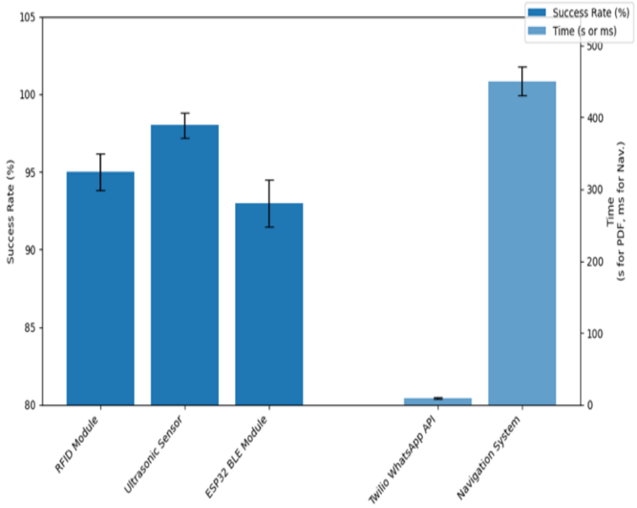

Fig. 8. Real Time System Performance Metrics (mean ± SD, n=10)

These results confirm that the integrated ML-driven trolley system is stable, responsive, and operationally efficient, proving its suitability for deployment in modern retail environments [3,7,9].

V. Conclusion and Future Research

A. Conclusion

An automated trolley system powered by ML optimization was described in this paper for enhancing item retrieval and product navigation alongside customer engagement in present-day retail outlets. RFID system works with ultrasonic sensors for obstacle spotting while BLE tracks customers for proximity functions. The system relies on Raspberry Pi 4B microcontrollers to execute autonomous navigation and process real-time data information with accuracy.The system integrates two recommendation algorithms: Apriori for association rule mining along with Random Forest for classification in its core functionality. The system combines these two algorithms effectively since both provide power to create precise individualized recommendations. The Random Forest model demonstrated top-level classification results to strengthen the hybrid system which generated recommendations with accuracy reaching up to 94%.

Checkout is fully automated through the generation of a PDF invoice, which is instantly delivered to the customer's WhatsApp number via the Twilio WhatsApp API assisted with Firebase cloud storage, replacing traditional GSM-based SMS alerts. This ensures a contactless, transparent, and seamless billing process, significantly improving the overall customer experience. Key experimental findings include:

- High accuracy and responsiveness across all hardware components.
- Effective real-time obstacle detection and autonomous movement.
- Robust customer tracking using BLE.
- Accurate and personalized product recommendations through the hybrid ML model.
- Instant digital billing via WhatsApp, improving operational efficiency and user convenience.

B. Future Research

The upcoming activities of the research will revolve around developing a system that is capable of managing larger and more complex retail businesses. Such improvements include the following:

- The integration of additional sensors within the IoT environment would make the system capable of better context understanding.
- Further development of the recommendation model should be based on improving both speed and accuracy of predictions.
- Developing user-oriented interfaces, such as an app or a dashboard, for real-time cart management and analysis.
- Actual deployment in retail locations to collect feedback from users to facilitate future system improvements.

The suggested approach is found to become a very efficient system that would change the functioning of retail logistics by making the process fully automated with the help of smart analytics and communication [14][15]. In addition, by providing IoT device and Machine Learning capabilities to create an interactive digital shopping environment, this solution provides a clear contribution towards fulfilling the UN SDGs. Specifically, by developing the technological capabilities of industry 4.0 within retail; supporting SDG 9, through the use of data analytics to improve performance and achieving SDG 12 by providing more effective stock tracking and management, reducing waste to achieve sustainable outcomes via responsible consumption.

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